

Introduction

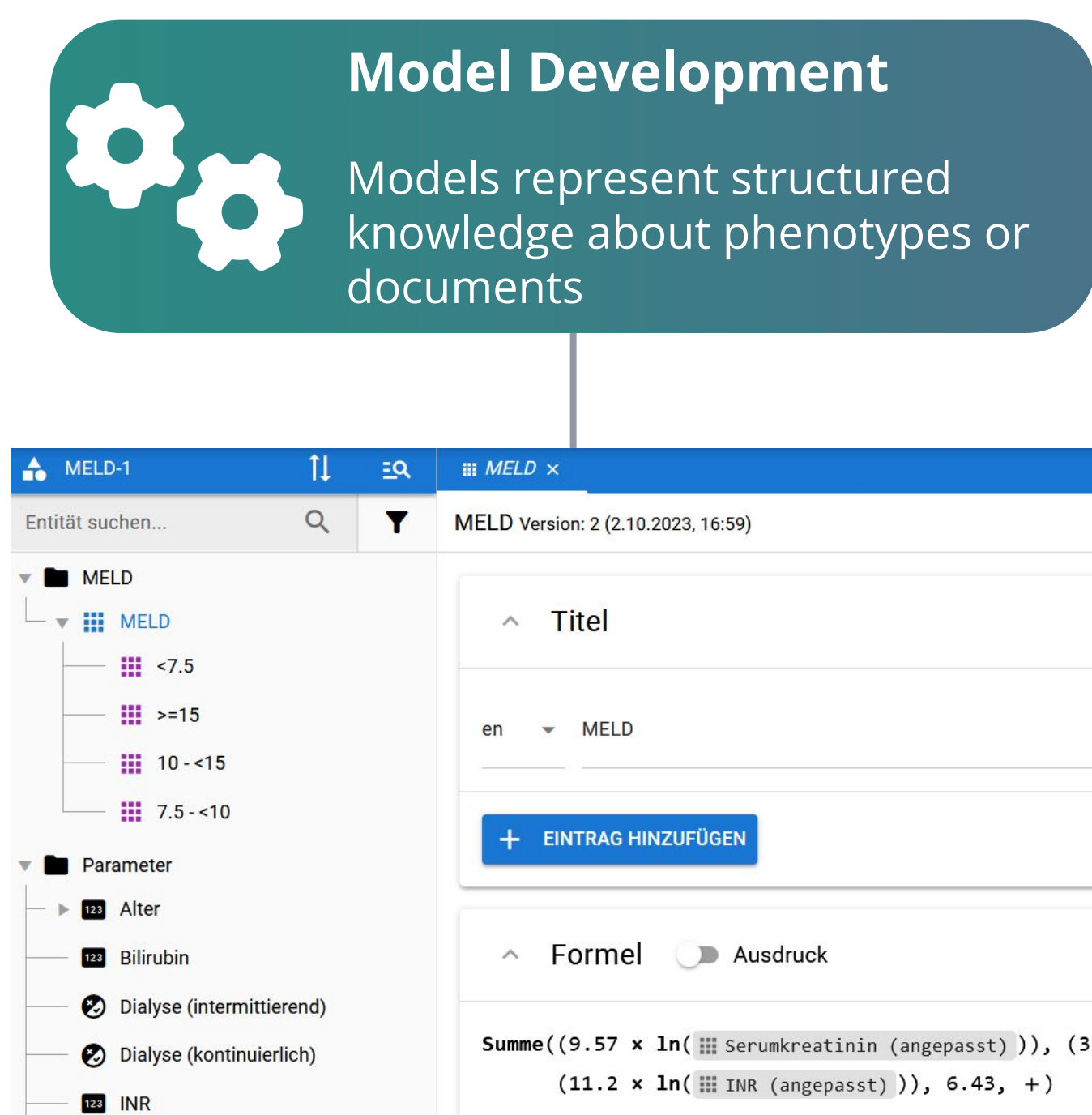
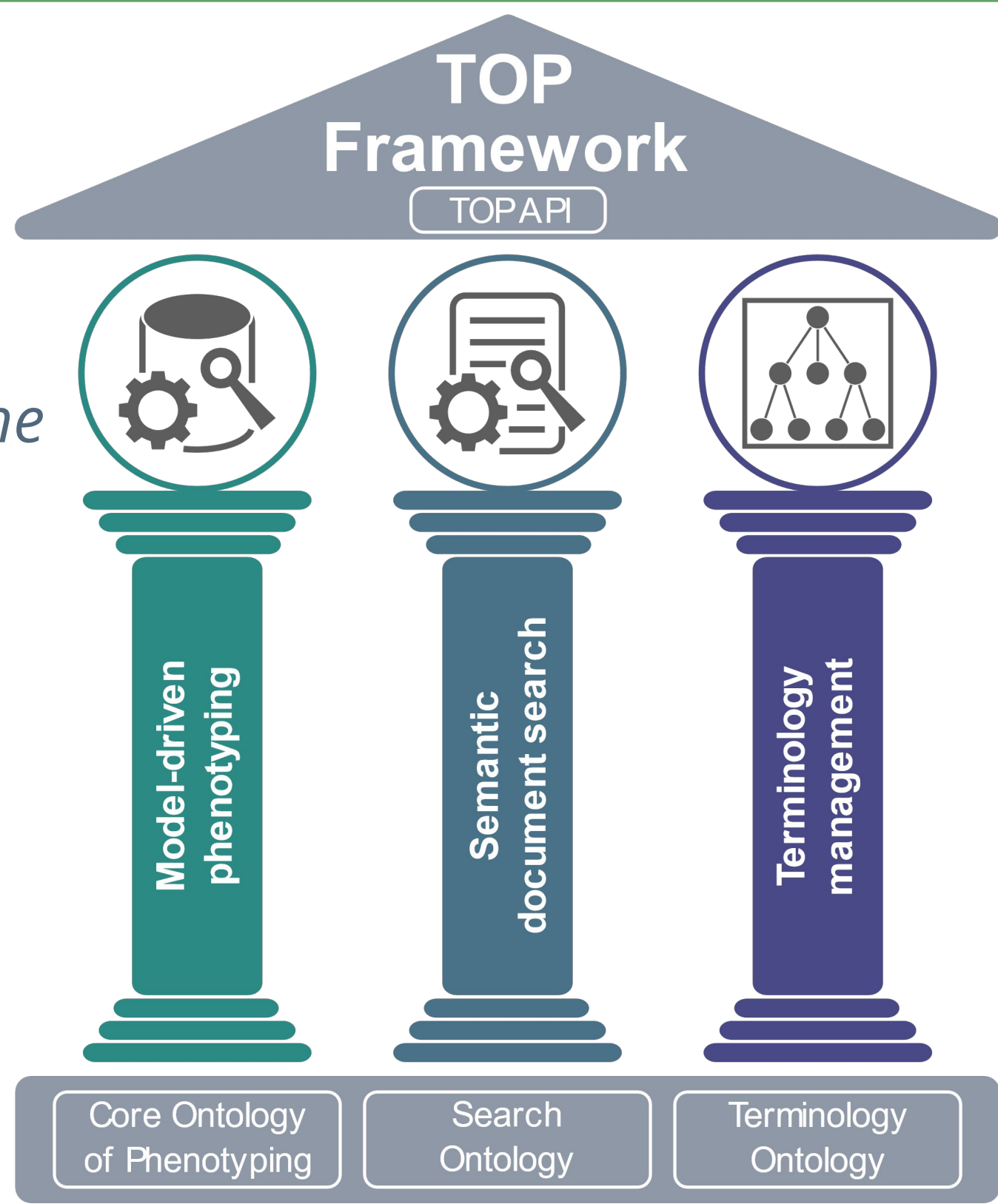
Next-generation phenotyping combines the **semantic and structured (standardized)** definition of phenotypes with the **creation of phenotype algorithms**. – Mo et al. [6].

Aims of the TOP Project

1. Ontological foundation of phenotyping [1]
2. Standardised phenotype modelling language
3. Combination of phenotype models with data sources in the MII, enabling search queries [2]
4. Integration of external terminologies [3]
5. Document search in unstructured documents [4]
6. User-friendly tools for next-generation phenotyping [5]

Result 1: Model-driven Phenotyping

- Phenotypes are observable (combinations of characteristics of an organism)
 - ↳ Basic Phenotype Class
 - ↳ Derived Phenotype Class
- Purposes:
 - Detection of diseases
 - Risk assessment
 - Recruitment for clinical studies
 - Individualised patient care and treatment



Phenotype model for end-stage liver disease (MELD).

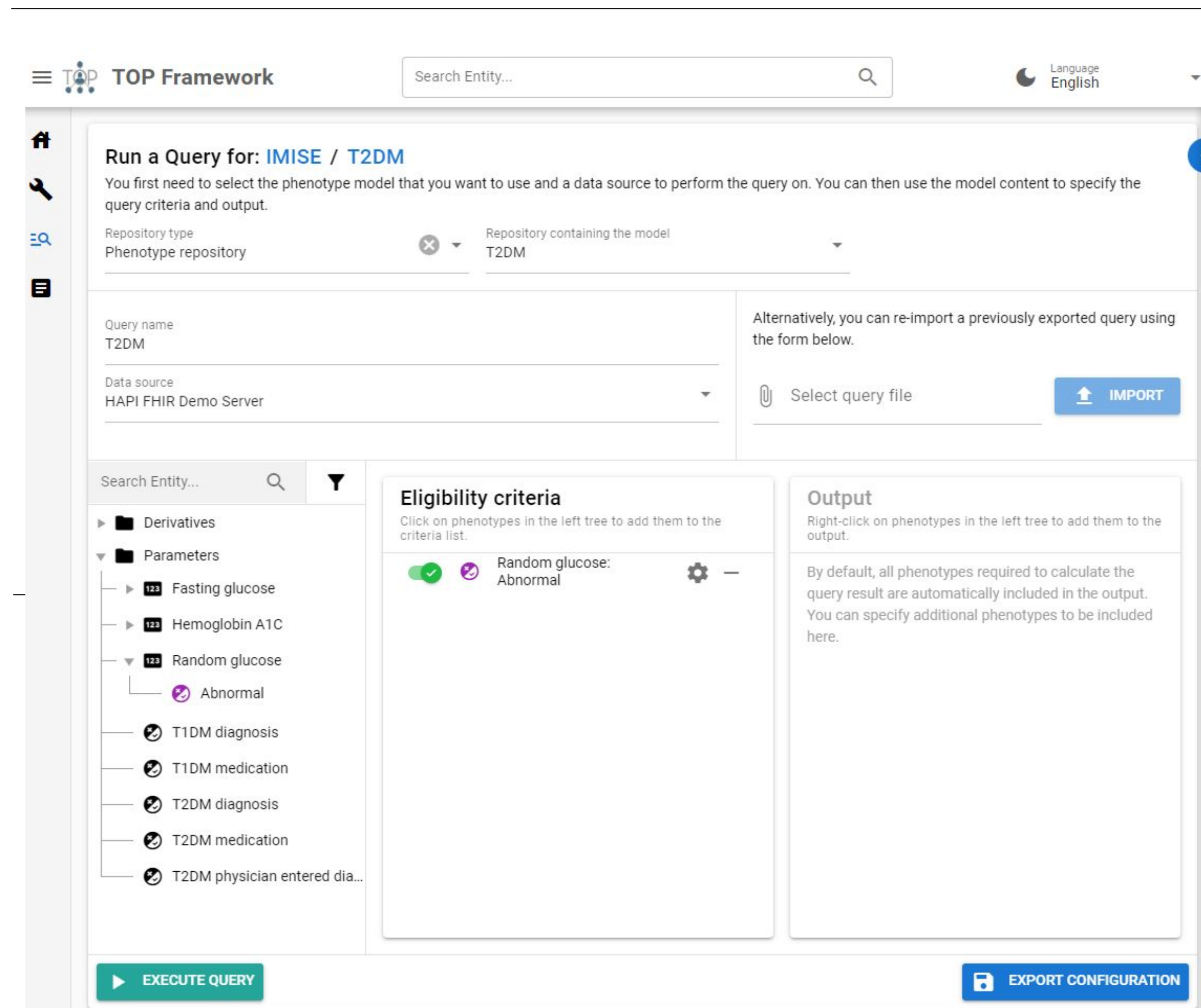
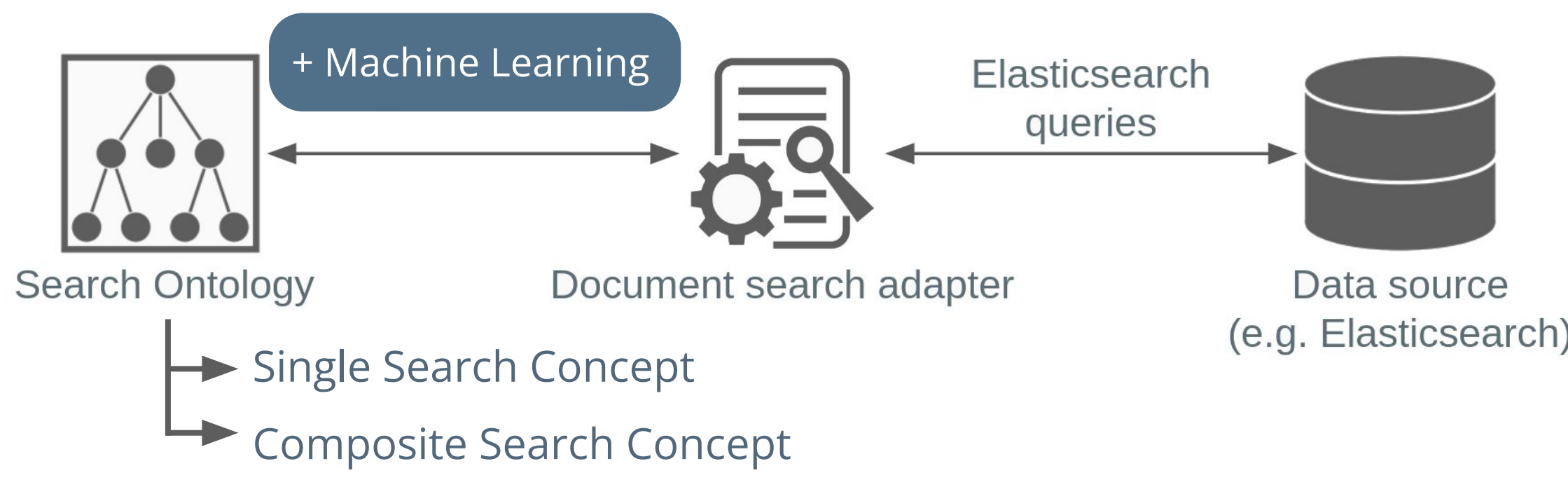
Methods

Formal ontologies provide a standardized structural and semantic foundation for the different tasks:

- Core Ontology of Phenotyping for model-driven phenotyping
- Search Ontology for semantic document search
- Terminology Ontology for managing the integration of external biomedical terminologies

Result 2: Document Search

- Unstructured documents are represented by concept graphs [7]
- Search ontologies capture complex query parameters, such as synonyms and subterms



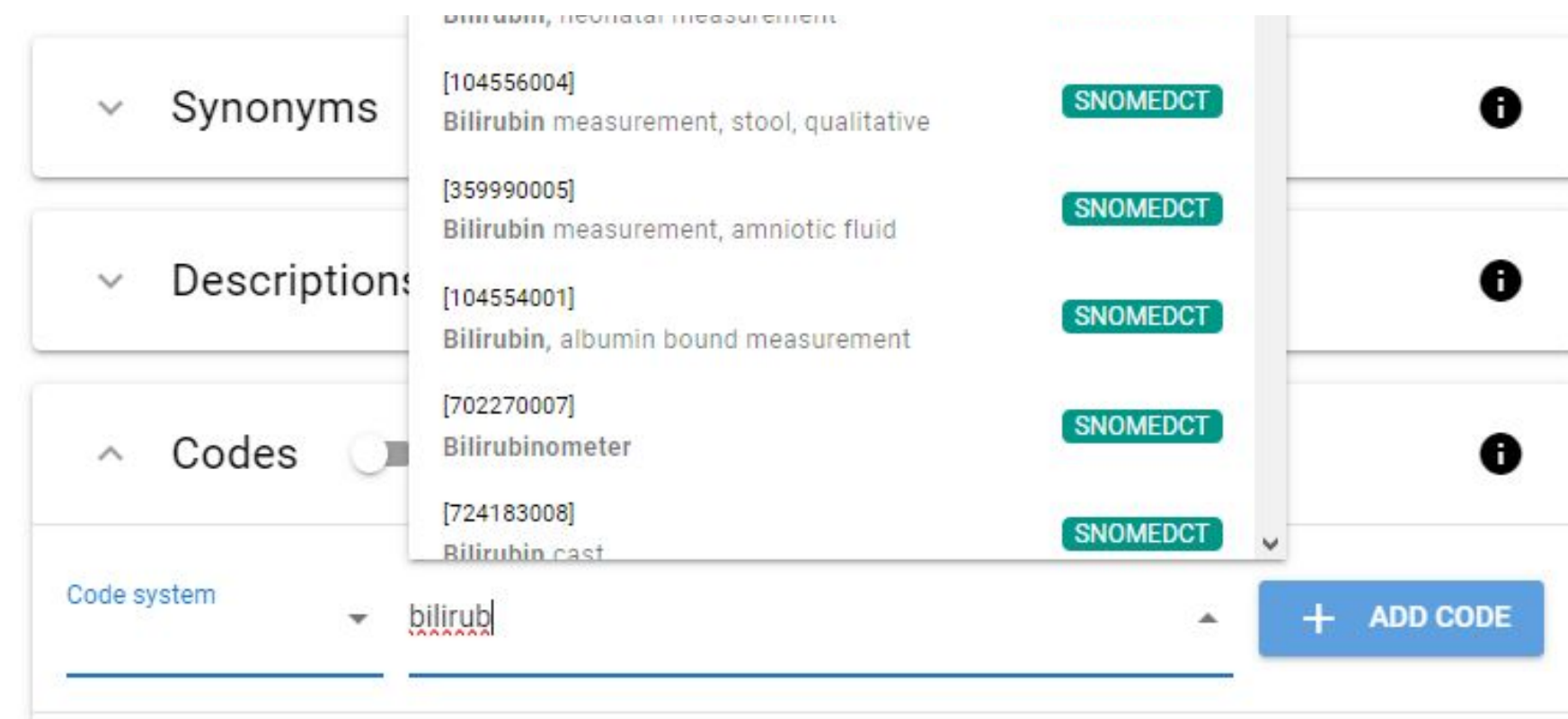
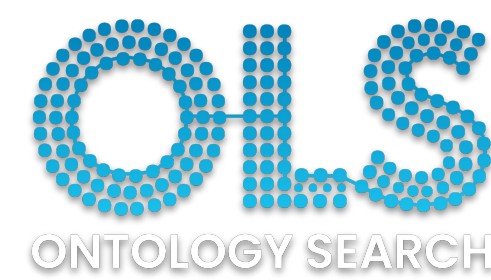
Exemplary phenotype search query.

Result 3: Terminology Management

- TOP Terminology Ontology allows unified access to external terminologies
- Semi-automated versioning approach keeps track of lexical and semantic changes in terminologies and ensures inter-version compatibility [3]

Integration of Terminologies

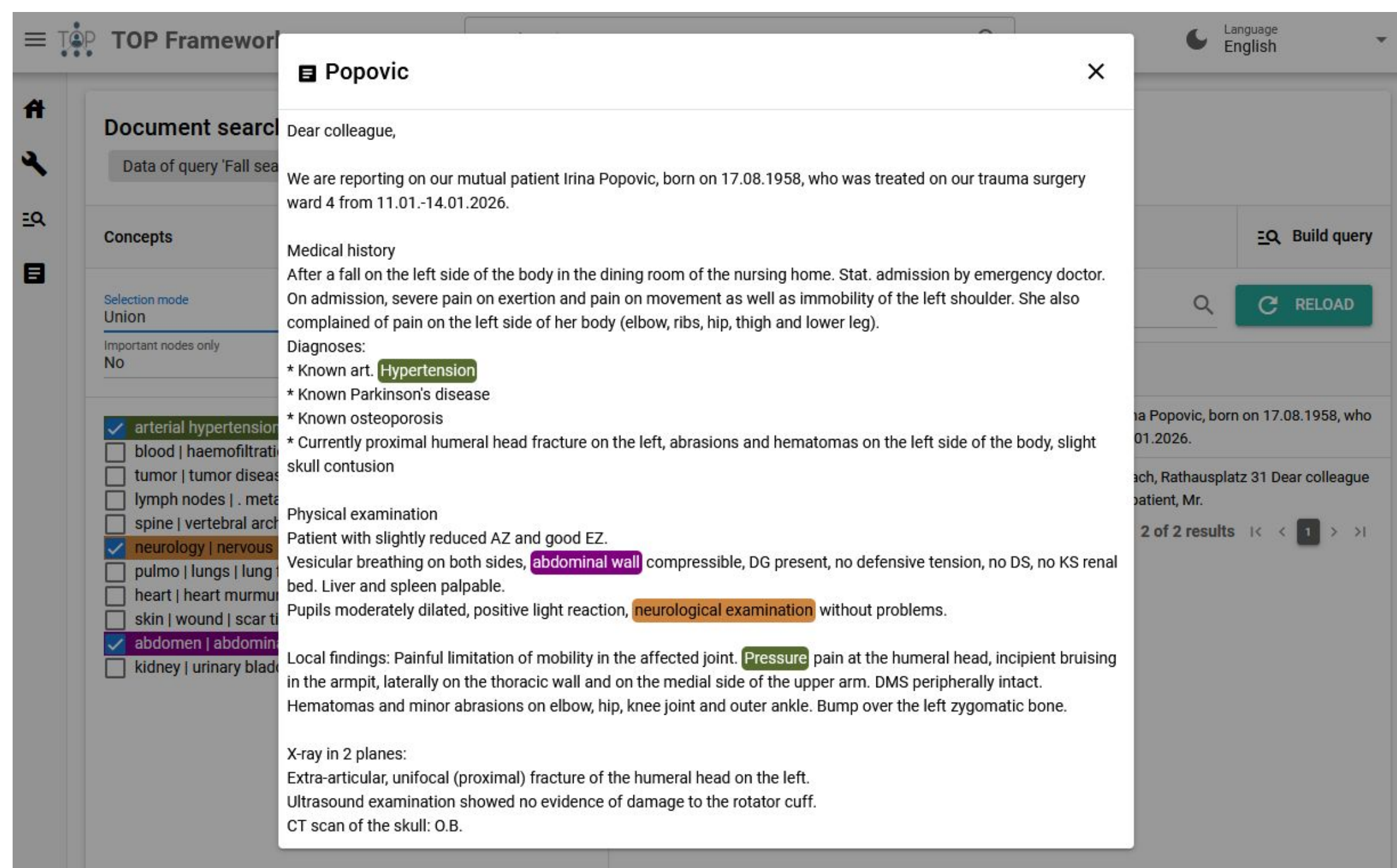
- Mapping of phenotypes and concepts to codes
- Adapters for external terminology servers (e.g., OLS)
- Import/versioning pipelines



Terminology server using the Ontology Lookup Service (OLS).

Search Queries

- Models are translated into query languages
- Data sources are accessible via adapters:
- generic adapters with configurations (e.g., OMOP, FHIR Search)
- specific adapters (e.g., LIFE study)



Exemplary documents from GraSCCo (Graz Synthetic Clinical Corpus).

Conclusion

- The TOP Framework, built upon the COP and TOP API, empowers researchers with **standardised models** and facilitates **efficient data exploration and analysis**.
- **Integration into the MII infrastructure** for streamlined data access.
- **Support of clinical research** by enhancing efficiency, collaboration, and knowledge discovery.



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Fotograf: Swen Reichhold

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